

Report

SOC THREAT DETECTION IN A SEGMENTED HEALTHCARE NETWORK

ORGANIZATION:

MediCore Health Systems

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Role: Security Operations Center (SOC) Analyst

Tools Used: Draw.io, pfSense, Wazuh, Ubuntu Server, Kali Linux, Wireshark

July 15th, 2026

TABLE OF CONTENTS

1. Executive Summary

2. Project Introduction

3. Methodology

4. Phase-by-Phase Analysis

- ○ Phase 1: Network Design & Segmentation (Draw.io & pfSense)
- ○ Phase 2: Attack Simulation & Traffic Capture (Kali Linux & Wireshark)
- ○ Phase 3: Security Event Monitoring & Response (Wazuh SIEM)

5. Final Findings & Impact

6. Recommendations

7. Conclusion

8. References

9. Appendices

- ○ Appendix A — Network Design & Firewall Configuration
- ○ Appendix B — Attack Simulation & Traffic Analysis
- ○ Appendix C — Wazuh SIEM Detection & Response

10. Key Analytical Questions

EXECUTIVE SUMMARY

MediCore Health Systems, a growing healthcare provider, faced significant risks due to a lack of internal network boundaries, exposing critical Electronic Health Records (EHR) and medical devices to potential lateral threats.

To address this, I performed a three-phase SOC analysis on MediCore network utilizing Wireshark, pfSense, and Wazuh. The objective was to identify and address network anomalies, block harmful traffic, and investigate possible threats. I identified and mitigated several critical indicators of compromise (IoCs) and provided final recommendations to enhance the organization's cybersecurity stance, a secure infrastructure was engineered using *Draw.io* for design and *pfSense* for implementing a three-zone segmentation strategy (WAN, LAN, and DMZ). Security visibility was established by deploying *Wazuh SIEM* to monitor clinical and administrative endpoints, while *Wireshark* was utilized for deep packet inspection.

Simulated attacks, including credential-based threats and internal reconnaissance from a *Kali Linux* platform, were successfully identified and analyzed through centralized logging. The investigation confirmed that without these controls, the organization remains vulnerable to data breaches and HIPAA non-compliance. Primary recommendations include hardening internal access policies, refining SIEM alert baselines, and implementing continuous monitoring to ensure the integrity of patient care and data.

2. PROJECT INTRODUCTION

Organization Overview: MediCore Health Systems is a healthcare provider managing Electronic Health Records (EHR), patient portals, telehealth services, and connected medical devices (IoMT).

- ***Security Concerns:*** Recent internal assessments identified unusual traffic originating from internal workstations targeting the EHR database. Gaps included a lack of network segmentation, weak access controls, and limited visibility into lateral movement, threatening HIPAA compliance and patient safety.
- ***Objectives:*** The project seeks to design a secure enterprise architecture, implement firewall-driven segmentation, establish centralized security monitoring, and validate these defenses through simulated attack and response workflows.

MediCore Health Systems recently observed a rise in network anomalies that raised alarms about malware, unauthorized access, and insider threats. As a member of the SOC team, I was responsible for deploying security tools, capturing traffic, and responding to incidents using practical methods. This report details the approach, implementation, and outcomes.

3. METHODOLOGY

Network Design: Utilize *Draw.io* to document a topology featuring isolated security zones (WAN, LAN, DMZ) and defined IP addressing.

- ***pfSense Configuration:*** Deploy pfSense to act as the security layer, configuring interfaces and firewall rules to regulate traffic between zones.
- ***Wazuh Deployment:*** I Install the *Wazuh Manager* on a dedicated server and deploy *Wazuh Agents* on DMZ and LAN endpoints for log collection and event correlation.

- **Attack Simulation:** I use *Kali Linux* to execute reconnaissance (Nmap scanning) and credential-based attacks against the infrastructure.
- **Traffic Capture:** I also employ *Wireshark* at strategic points to capture and analyze malicious packet data and identify Indicators of Compromise (IoCs).
- **Detection & Response:** I correlate Wazuh alerts with packet captures to investigate incidents and apply hardening measures such as updated firewall rules.

4. PHASE-BY-PHASE ANALYSIS

PHASE 1: Network Design & Segmentation

- **Objective:** Design a segmented network and implement access controls.
- **Key Actions:** Documented topology in Draw.io; configured pfSense to separate clinical assets from patient-facing services.
- **Expected Security Controls:** Isolation of EHR systems, restricted DMZ access, and defined routing policies.

The screenshot displays the pfSense web interface. At the top, there's a navigation bar with links to 'Status / Dashboard' and a '+ ?' icon. Below this, the 'System Information' panel is visible on the left, and the 'Netgate Services And Support' panel is on the right. The 'System Information' panel shows details about the system, including the name 'MediCore-Health.home.arpa', user 'admin@192.168.60.11', system 'VirtualBox Virtual Machine', BIOS 'Innotek GmbH', version '2.8.1-RELEASE (amd64)', CPU type '13th Gen Intel(R) Core(TM) i7-1355U', hardware crypto 'Inactive', kernel PTI 'Disabled', MDS Mitigation 'Inactive', uptime '00 Hour 38 Minutes 36 Seconds', current date/time 'Sun Jul 12 8:36:57 UTC 2026', and DNS server(s) '127.0.0.1', '::1', and '10.225.42.125'. The 'Netgate Services And Support' panel shows the contract type 'Community Support' and provides links to various support resources. At the bottom, the 'Interfaces' panel lists three interfaces: WAN (1000baseT <full-duplex> 192.168.50.4), LAN (1000baseT <full-duplex> 192.168.60.1), and DMZ (1000baseT <full-duplex> 192.168.70.1).

System Information	
Name	MediCore-Health.home.arpa
User	admin@192.168.60.11 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: 04d59c9fa69db2df82da
BIOS	Vendor: Innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Wed May 13 19:43:00 UTC 2026 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Sun Jul 12 7:59:38 UTC 2026
CPU Type	13th Gen Intel(R) Core(TM) i7-1355U 2 CPUs: 1 package(s) x 2 cache groups x 1 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 38 Minutes 36 Seconds
Current date/time	Sun Jul 12 8:36:57 UTC 2026
DNS server(s)	127.0.0.1 ::1 10.225.42.125

Netgate Services And Support	
Contract Type	Community Support Community Support Only
NETGATE & PFSense COMMUNITY SUPPORT RESOURCES If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY. You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space. - Upgrade Your Support - Netgate Global Support FAQ - Netgate Professional Services - Community Support Resources - Official pfSense Training by Netgate - Visit Netgate.com If you decide to purchase a Netgate Global TAC Support subscription, you MUST have your Netgate Device ID (NDI) from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase a TAC support plan here.	

Interfaces	
WAN	1000baseT <full-duplex> 192.168.50.4
LAN	1000baseT <full-duplex> 192.168.60.1
DMZ	1000baseT <full-duplex> 192.168.70.1

192.168.50.4/firewall_rules.php
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
pfSense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help

WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Firewall / Rules / WAN

Floating
WAN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/1.12 MiB	*	*	*	WAN Address	443 80	*	*	*	Anti-Logout Rule	⚙
✗	0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*	*	Block bogus networks	⚙

No rules are currently defined for this interface
All incoming connections on this interface will be blocked until pass rules are added. Click the button to add a new rule.

Add
Add
Delete
Toggle
Copy
Save
Separator

Draw.io | Enterprise Topology, Security Zones & IP Addressing

Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete
FreeBSD/amd64 (MediCore-Health.home.arpa) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: 04d59c9fa69db2df82da
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on MediCore-Health ***
WAN (wan) -> em0 -> v4/DHCP4: 192.168.50.4/24
LAN (lan) -> em1 -> v4: 192.168.60.1/24
DMZ (opt1) -> em2 -> v4: 192.168.70.1/24
0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM
Enter an option:

PHASE 2: Attack Simulation & Traffic Capture

- **Objective:** Simulate realistic attacks to test visibility.
- **Key Actions:** Conducted port scans and authentication attacks from Kali Linux; captured traffic in Wireshark to analyze protocol behavior.
- **Observed Results:** Identification of specific IP sources and methods used during the simulated breach.

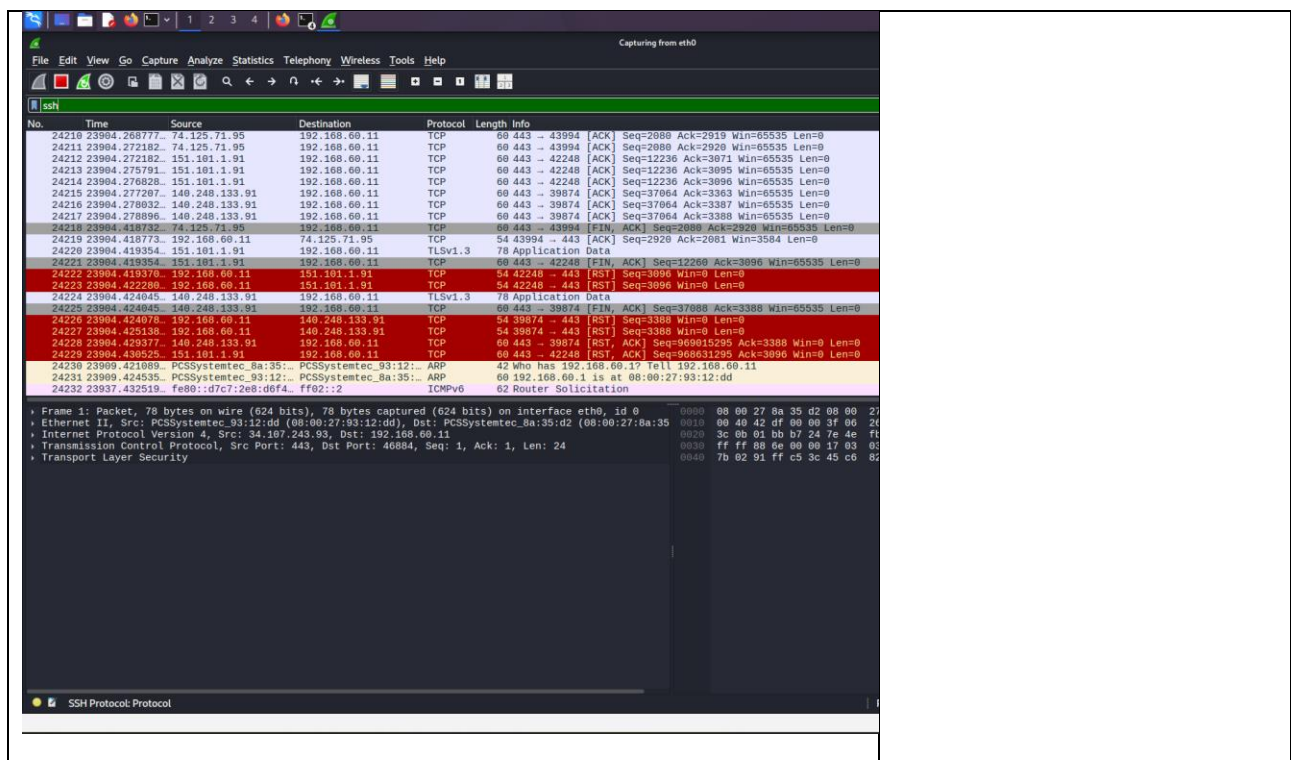
```
(kali@kali)~$ nmap -sU 192.168.20.1 -p 161 --script=snmp-interfaces 192.168.60.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 06:39 -0400
Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)
Host is up (0.010s latency).

PORT      STATE SERVICE
161/udp   open  snmp
| snmp-interfaces:
|   em0
|     IP address: 192.168.50.4 Netmask: 255.255.255.0
|     MAC address: 08:00:27:c3:63:ee (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 17.98 Mb sent, 758.18 Mb received
|   em1
|     IP address: 192.168.60.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:93:12:dd (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 758.14 Mb sent, 20.98 Mb received
|   em2
|     IP address: 192.168.70.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:f8:5c:13 (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 3.16 Mb sent, 2.72 Mb received
|   enc0
|     Type: wlanPP2 Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   lo0
|     IP address: 127.0.0.1 Netmask: 255.0.0.0
|     Type: softwareLoopback Speed: 0 Kbps
|     Status: up
|     Traffic stats: 38.44 Kb sent, 38.44 Kb received
|   pflog0
|     Type: ifPwType Speed: 0 Kbps
|     Status: down
|     Traffic stats: 40.84 Kb sent, 0.00 Kb received
|   pfsync0
|     Type: ilan Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)
Nmap done: 2 IP addresses (1 host up) scanned in 5.45 seconds
```

```
(kali@kali)~$ hydra -l users.txt -P password.txt -t 4 -V ssh://192.168.70.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-14 04:40:53
[DATA] max 4 tasks per 1 server, overall 4 tasks, 25 login tries (1:5/p:5), ~7 tries per task
[DATA] attacking ssh://192.168.70.10:22/
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 1 of 25 [child 0] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 2 of 25 [child 1] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 3 of 25 [child 2] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 4 of 25 [child 3] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 5 of 25 [child 2] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 6 of 25 [child 0] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 7 of 25 [child 3] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 8 of 25 [child 2] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 9 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: admin password: 9090
[ATTEMPT] target 192.168.70.10 - login "buntu" - pass "9090" - 11 of 25 [child 0] (0/0)
[22][ssh] host: 192.168.70.10 login: admin password: 9090
[ATTEMPT] target 192.168.70.10 - login "buntu" - pass "9090" - 12 of 25 [child 2] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 13 of 25 [child 3] (0/0)
[22][ssh] host: 192.168.70.10 login: admin password: 9090
[ATTEMPT] target 192.168.70.10 - login "buntu" - pass "9090" - 14 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 16 of 25 [child 0] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 17 of 25 [child 3] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 18 of 25 [child 2] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 19 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 21 of 25 [child 0] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 22 of 25 [child 3] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 23 of 25 [child 2] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 24 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: ApexTrust password: 9090
[22][ssh] host: 192.168.70.10 login: ApexTrust password: 9090
[STATUS] 25.00 tries/min, 25 tries in 00:01h, 1 to do in 00:01h, 2 active
[22][ssh] host: 192.168.70.10 login: ApexTrust password: 9090
1 of 1 target successfully completed, 16 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-14 04:42:00
```

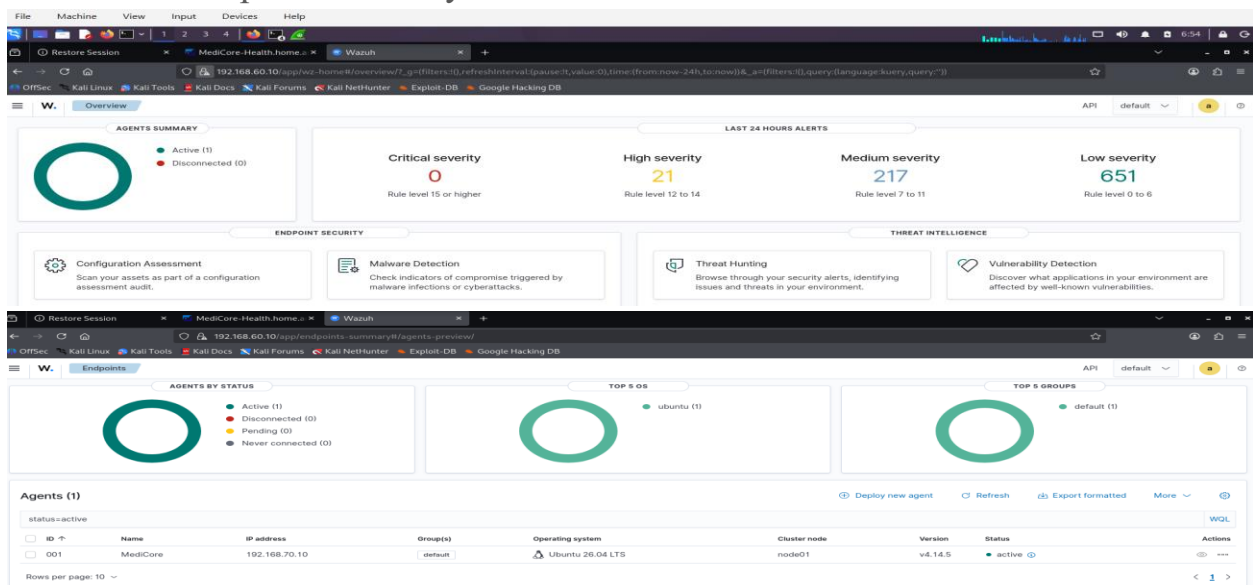
```
Session Actions Edit View Help
L-$ nmap 192.168.60.11/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-13 14:44 -0400
Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)
Host is up (0.020s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp    open  https
| MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)
|   Active [1]
Nmap scan report for 192.168.60.10
Host is up (0.0056s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp    open  https
| MAC Address: 08:00:27:94:5D:6B (Oracle VirtualBox virtual NIC)
|   Disconnected [0]
|   Pending [0]
|   Server-Disconnected [0]
Nmap scan report for 192.168.60.11
Host is up (0.028s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
Nmap done: 256 IP addresses (3 hosts up) scanned in 19.42 seconds
(kali@kali)~$ nmap -sV 192.168.60.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-13 14:59 -0400
Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)
Host is up (0.0094s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain Unbound
80/tcp    open  http  nginx
443/tcp    open  ssl/http nginx
| MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)
Service detection performed. Please report any incorrect results
Nmap done: 1 IP address (1 host up) scanned in 18.63 seconds
(kali@kali)~$ nmap -sU 192.168.60.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-13 15:15 -0400
Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)
Host is up (0.017s latency).
Not shown: 997 open/filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  ntp
161/udp   open  snmp
| MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 9.28 seconds
```

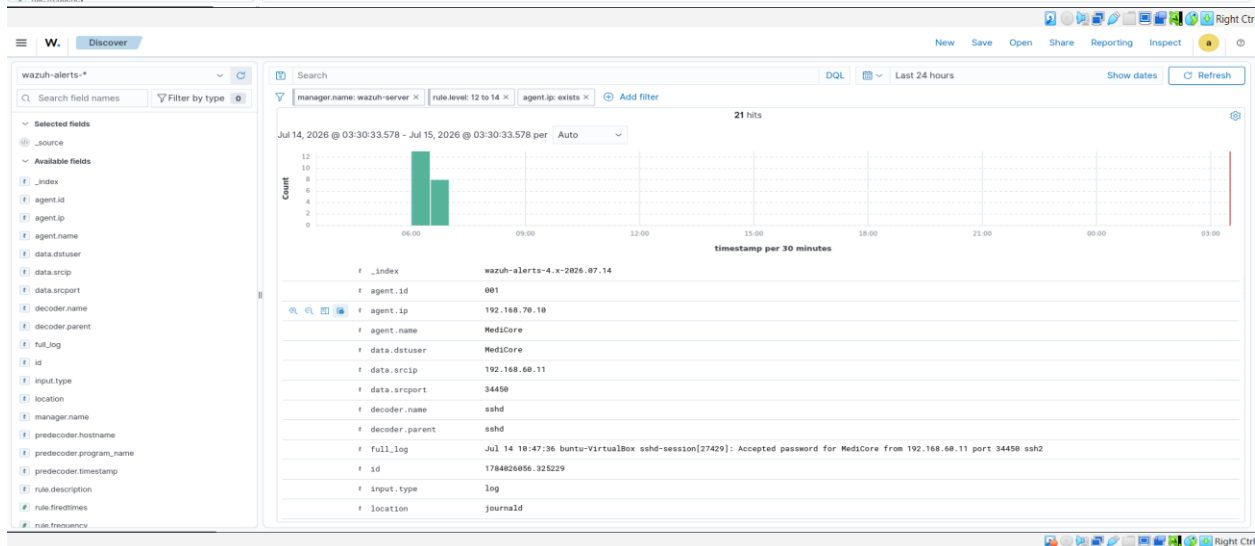
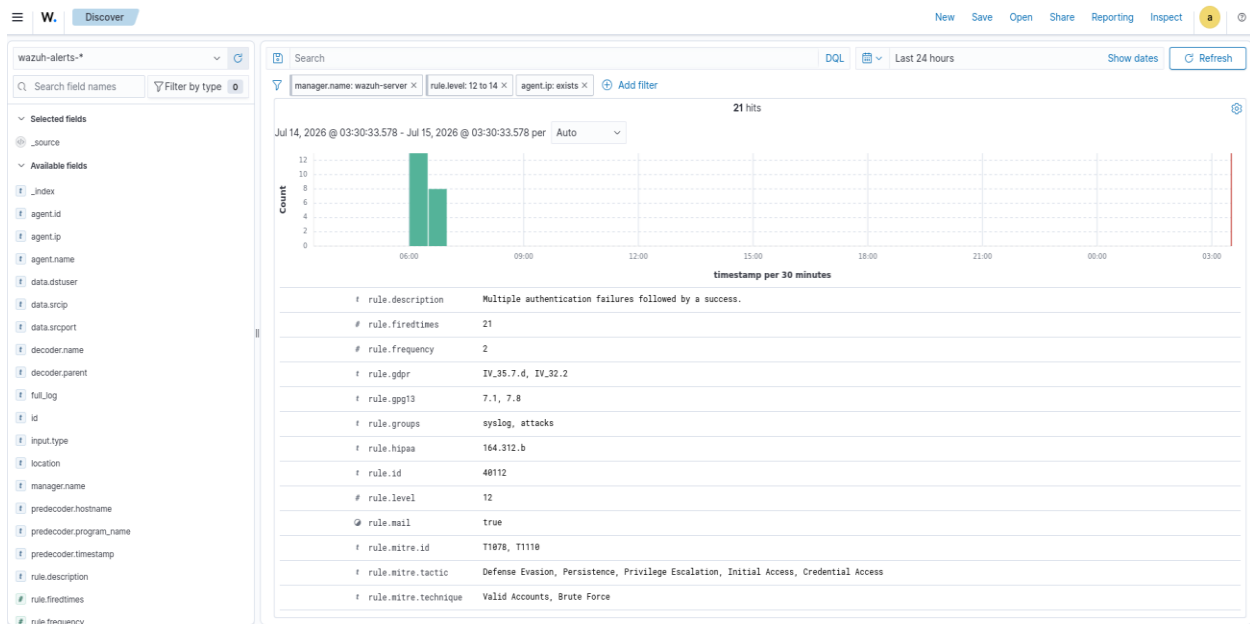
```
(kali@kali)~$ nmap -sV 192.168.60.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-13 14:59 -04
Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)
Host is up (0.0094s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain Unbound
80/tcp    open  http  nginx
443/tcp    open  ssl/http nginx
| MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)
Service detection performed. Please report any incorrect result
Nmap done: 1 IP address (1 host up) scanned in 18.63 seconds
```



PHASE 3: Security Event Monitoring & Response

- **Objective:** Correlate logs and perform incident response.
- **Key Actions:** I Monitored the Wazuh dashboard for real-time alerts; analyzed timestamps and severity of events to build a timeline of the attack.





5. FINAL FINDINGS & IMPACT

- **Vulnerabilities:** Overly permissive legacy firewall rules and a lack of monitoring for internal-to-internal traffic.
- **Impact:** Potential for unauthorized access to Protected Health Information (PHI), leading to massive fines, loss of patient trust, and clinical disruption.
- **IoCs:** Specific malicious IP addresses (from the Kali instance) and high-frequency authentication failure logs.

6. RECOMMENDATIONS

- **Technical Controls:** To Implement "Least Privilege" firewall rules and harden SSH/EHR server configurations.
- **Process Improvements:** Develop incident response playbooks for internal reconnaissance alerts.
- **Monitoring:** Tune Wazuh rules to reduce noise and establish a baseline for normal administrative traffic.
- **Long-term Strategy:** Regular patching of medical devices and ongoing SOC staff training on healthcare-specific threats.

7. CONCLUSION

This project capstone demonstrated that network segmentation is critical for protecting sensitive healthcare data. By simulating real-world SOC workflows from design to detection, the lab showed that integrated visibility through tools like Wazuh and pfSense is essential for containing threats before they impact patient care.

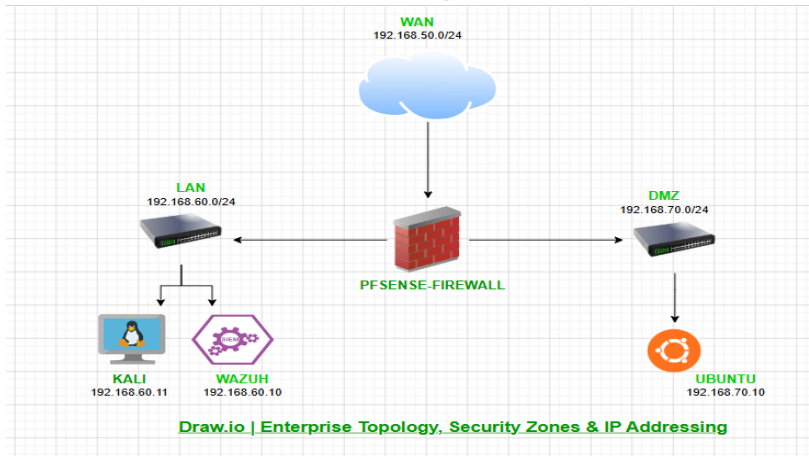
8. REFERENCES

- Wazuh Documentation <https://documentation.wazuh.com/>
- PfSenseManual-<https://docs.netgate.com/>
- Wireshark User Guide
- Kali Linux - <https://www.Kali.org/ Docs/>
- Nmap Documentation
- MediCore Health Systems Internal Security Brief

9. APPENDICES

Appendix A — Network Design & Firewall Configuration

9.1.a. Draw.io network diagram



9.2.a. pfSense interface screenshots and IP addressing table

```
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (MediCore-Health.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 04d59c9fa69db2df82da

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on MediCore-Health ***

WAN (wan)  -> em0 -> v4/DHCP4: 192.168.50.4/24
LAN (lan)  -> em1 -> v4: 192.168.60.1/24
DMZ (opt1) -> em2 -> v4: 192.168.70.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █
```


Firewall / Rules / DMZ

Floating

WAN

LAN

DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	2/6.32 MiB	IPv4 *	DMZ subnets	*	*	*	*	none		Anchor Edit Copy Delete Toggle

↑ Add

↓ Add

Delete

Toggle

Copy

Save

Separator

Firewall / Rules / LAN

Floating

WAN

LAN

DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/768 B	*	*	*	LAN Address	443 80	*	*	Anti-Lockout Rule	Settings
☐	✓	9/735.66 MiB	IPv4 *	LAN subnets	*	*	*	*	none	Default allow LAN to any rule	Anchor Edit Copy Delete Toggle
☐	✓	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none	Default allow LAN IPv6 to any rule	Anchor Edit Copy Delete Toggle

Firewall / Rules / WAN

Floating

WAN

LAN

DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗	0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*	Block bogon networks	Settings

Appendix B — Attack Simulation & Traffic Analysis

9.1.b. Kali Linux attack commands & screenshots

```

(kali@kali)-[~]
$ nmap -sU 192.168.20.1 -p 161 --script=snmp-interfaces 192.168.60.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 06:39 -0400
Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)
Host is up (0.010s latency).

PORT      STATE SERVICE
161/udp   open  snmp
| snmp-interfaces:
|   em0
|     IP address: 192.168.50.4 Netmask: 255.255.255.0
|     MAC address: 08:00:27:c3:63:ee (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 17.98 Mb sent, 758.18 Mb received
|   em1
|     IP address: 192.168.60.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:f8:5c:13 (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 758.14 Mb sent, 20.98 Mb received
|   em2
|     IP address: 192.168.70.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:f8:5c:13 (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 3.16 Mb sent, 2.72 Mb received
|   enc0
|     Type: wwanPP2 Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   lo0
|     IP address: 127.0.0.1 Netmask: 255.0.0.0
|     Type: softwareLoopback Speed: 0 Kbps
|     Status: up
|     Traffic stats: 38.44 Kb sent, 38.44 Kb received
|   pflog0
|     Type: ifPwType Speed: 0 Kbps
|     Status: down
|     Traffic stats: 40.84 Kb sent, 0.00 Kb received
|   pfsync0
|     Type: ilan Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)
Nmap done: 2 IP addresses (1 host up) scanned in 5.45 seconds

```

```

(kali@kali)-[~]
$ hydra -L users.txt -P password.txt -t 4 -V ssh://192.168.70.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-14 04:40:53
[DATA] max 4 tasks per 1 server, overall 4 tasks, 25 login tries (l:5/p:5), ~7 tries per task
[DATA] attacking ssh://192.168.70.10:22/
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 1 of 25 [child 0] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 2 of 25 [child 1] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 3 of 25 [child 2] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 4 of 25 [child 3] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 5 of 25 [child 2] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 6 of 25 [child 0] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 7 of 25 [child 3] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 8 of 25 [child 2] (0/0)
[ATTEMPT] target 192.168.70.10 - login "admin" - pass "9090" - 9 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: admin password: 9090
[ATTEMPT] target 192.168.70.10 - login "buntu" - pass "9090" - 11 of 25 [child 0] (0/0)
[22][ssh] host: 192.168.70.10 login: admin password: 9090
[ATTEMPT] target 192.168.70.10 - login "buntu" - pass "9090" - 12 of 25 [child 2] (0/0)
[22][ssh] host: 192.168.70.10 login: admin password: 9090
[ATTEMPT] target 192.168.70.10 - login "buntu" - pass "9090" - 13 of 25 [child 3] (0/0)
[22][ssh] host: 192.168.70.10 login: admin password: 9090
[ATTEMPT] target 192.168.70.10 - login "buntu" - pass "9090" - 14 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 16 of 25 [child 0] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 17 of 25 [child 3] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 18 of 25 [child 2] (0/0)
[22][ssh] host: 192.168.70.10 login: buntu password: 9090
[ATTEMPT] target 192.168.70.10 - login "MediCore" - pass "9090" - 19 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 21 of 25 [child 0] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 22 of 25 [child 3] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 23 of 25 [child 2] (0/0)
[22][ssh] host: 192.168.70.10 login: MediCore password: 9090
[ATTEMPT] target 192.168.70.10 - login "ApexTrust" - pass "9090" - 24 of 25 [child 1] (0/0)
[22][ssh] host: 192.168.70.10 login: ApexTrust password: 9090
[22][ssh] host: 192.168.70.10 login: ApexTrust password: 9090
[STATUS] 25.00 tries/min, 25 tries in 00:01h, 1 to do in 00:01h, 2 active
[22][ssh] host: 192.168.70.10 login: ApexTrust password: 9090
[22][ssh] host: 192.168.70.10 login: ApexTrust password: 9090
1 of 1 target successfully completed, 16 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-14 04:42:00

```

Session Actions Edit View Help

└─\$ nmap 192.168.60.11/24

Starting Nmap 7.98 (<https://nmap.org>) at 2026-07-13 14:44 -0400

Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)

Host is up (0.020s latency).

Not shown: 997 filtered tcp ports (no-response)

PORT STATE SERVICE

53/tcp open domain

80/tcp open http

443/tcp open https

MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.60.10

Host is up (0.0056s latency).

Not shown: 998 closed tcp ports (reset)

PORT STATE SERVICE

22/tcp open ssh

443/tcp open https

MAC Address: 08:00:27:94:5D:6B (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.60.11

Host is up (0.028s latency).

Not shown: 999 filtered tcp ports (no-response)

PORT STATE SERVICE

22/tcp open ssh

Nmap done: 256 IP addresses (3 hosts up) scanned in 19.42 seconds

└─(kali@kali)-[~]

└─\$ nmap -sV 192.168.60.1

Starting Nmap 7.98 (<https://nmap.org>) at 2026-07-13 14:59 -0400

Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)

Host is up (0.0094s latency).

Not shown: 997 filtered tcp ports (no-response)

PORT STATE SERVICE VERSION

53/tcp open domain Unbound

80/tcp open http nginx

443/tcp open ssl/http nginx

MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 18.63 seconds

└─(kali@kali)-[~]

└─\$ nmap -sU 192.168.60.1

Starting Nmap 7.98 (<https://nmap.org>) at 2026-07-13 15:15 -0400

Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)

Host is up (0.017s latency).

Not shown: 997 open|filtered udp ports (no-response)

PORT STATE SERVICE

53/udp open domain

123/udp open ntp

161/udp open snmp

MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.28 seconds

└─(kali@kali)-[~]

└─\$ nmap -sV 192.168.60.1

Starting Nmap 7.98 (<https://nmap.org>) at 2026-07-13 14:59 -0400

Nmap scan report for MediCore-Health.home.arpa (192.168.60.1)

Host is up (0.0094s latency).

Not shown: 997 filtered tcp ports (no-response)

PORT STATE SERVICE VERSION

53/tcp open domain Unbound

80/tcp open http nginx

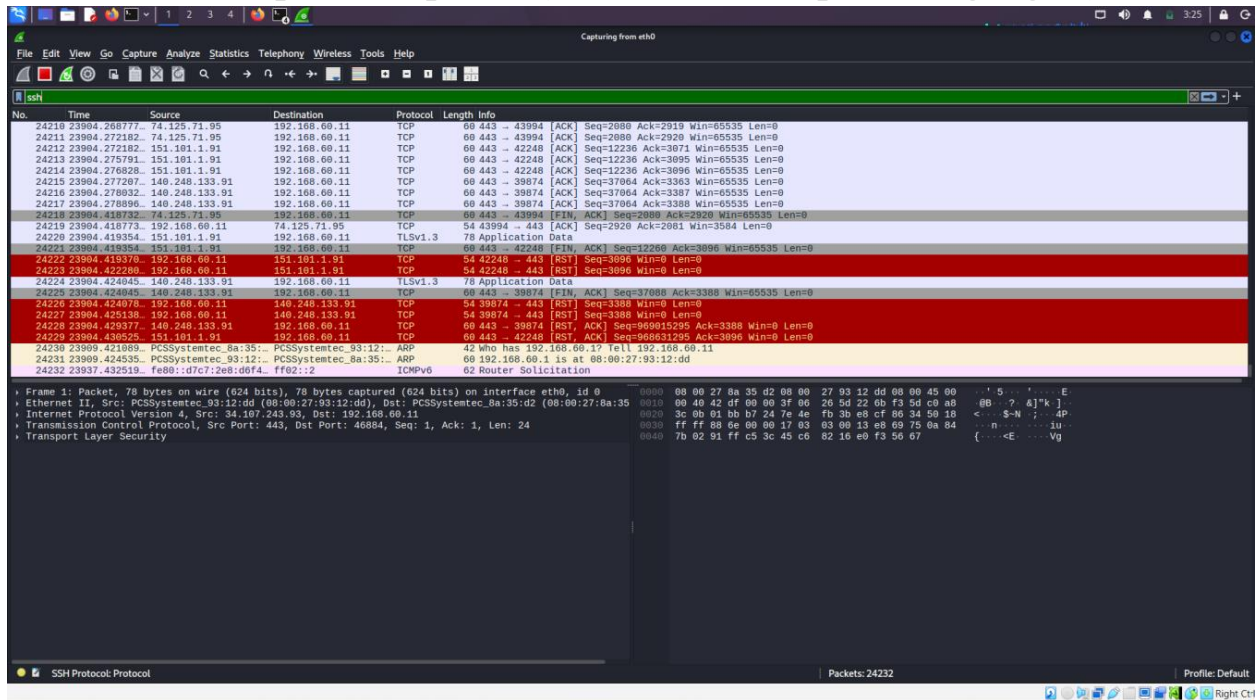
443/tcp open ssl/http nginx

MAC Address: 08:00:27:93:12:DD (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

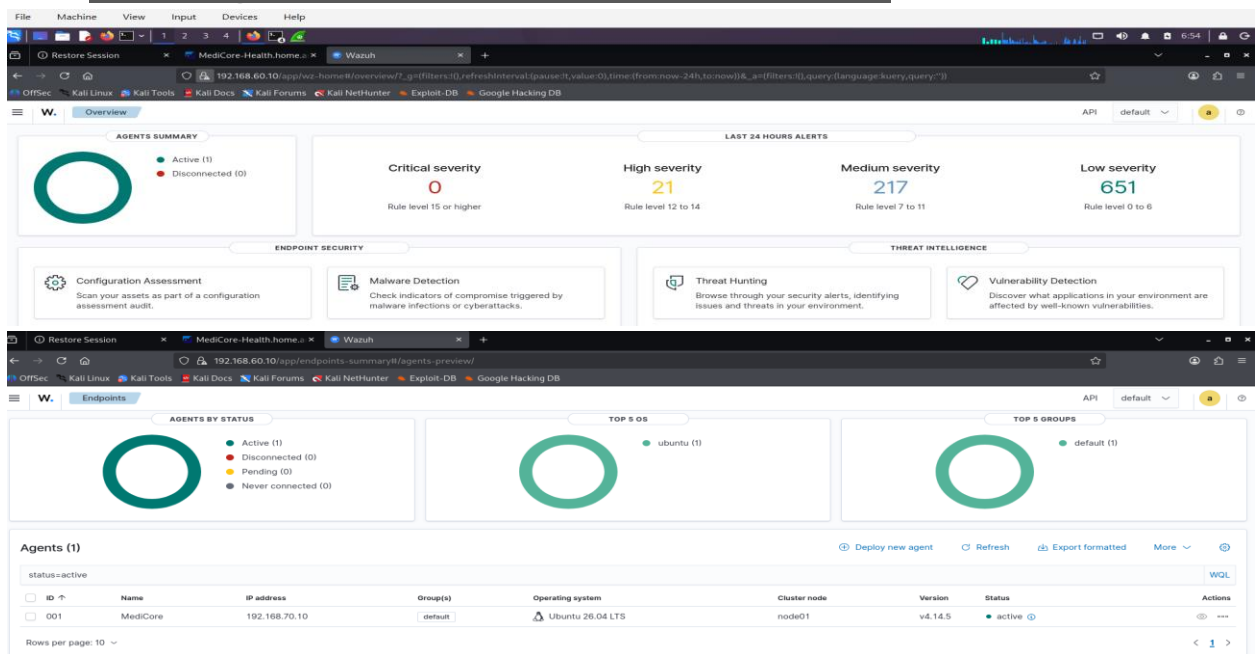
Nmap done: 1 IP address (1 host up) scanned in 18.63 seconds

9.2.b Wireshark packet captures (time, filter used, packet highlights)

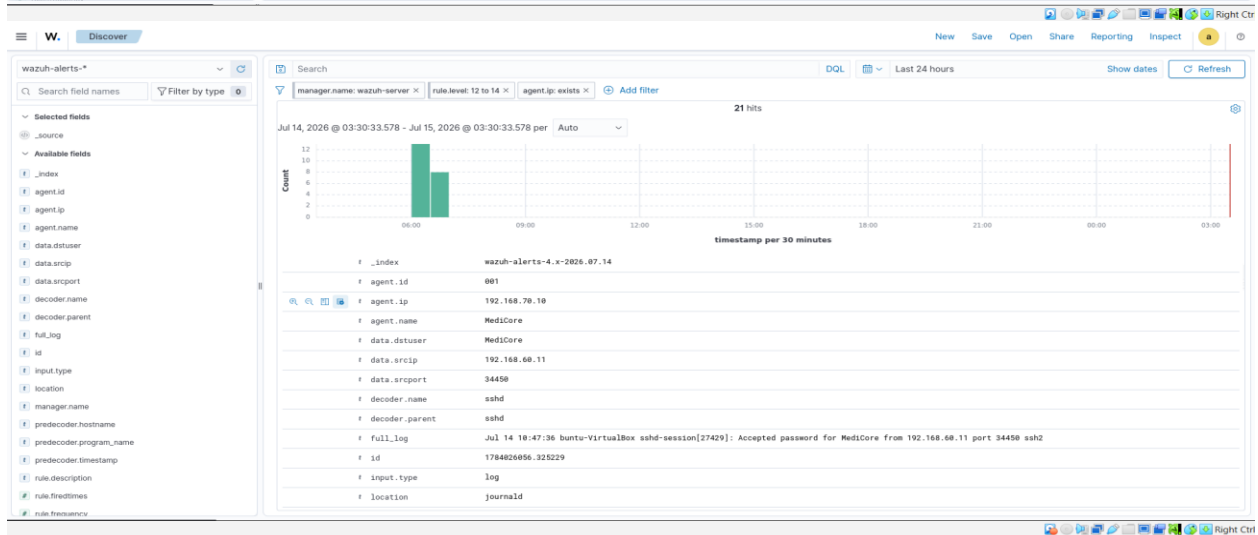
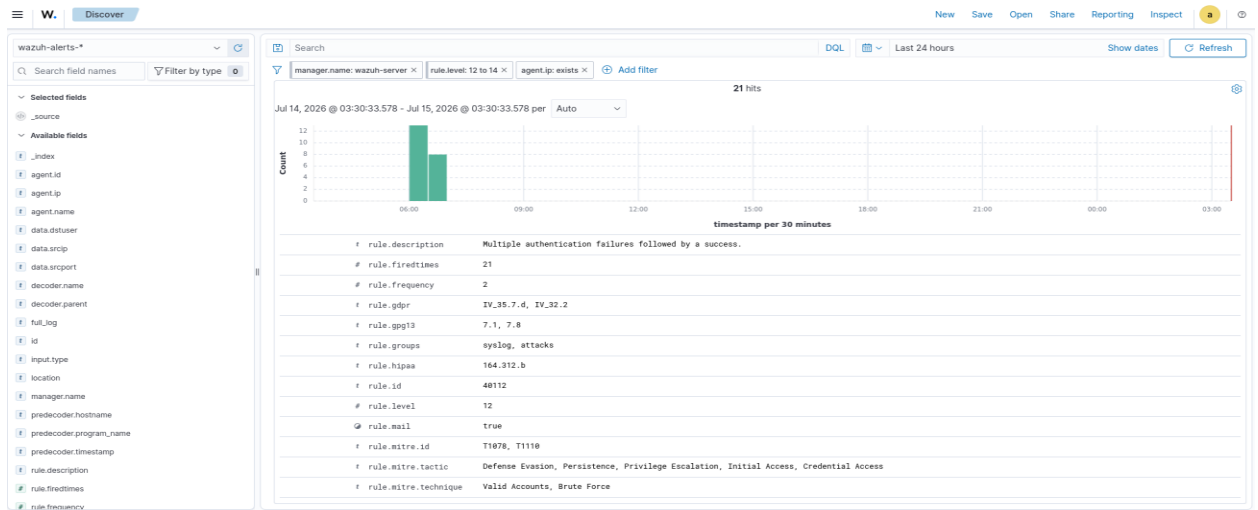


Appendix C — Wazuh SIEM Detection & Response

9.1.c Wazuh agent connection confirmation screenshots



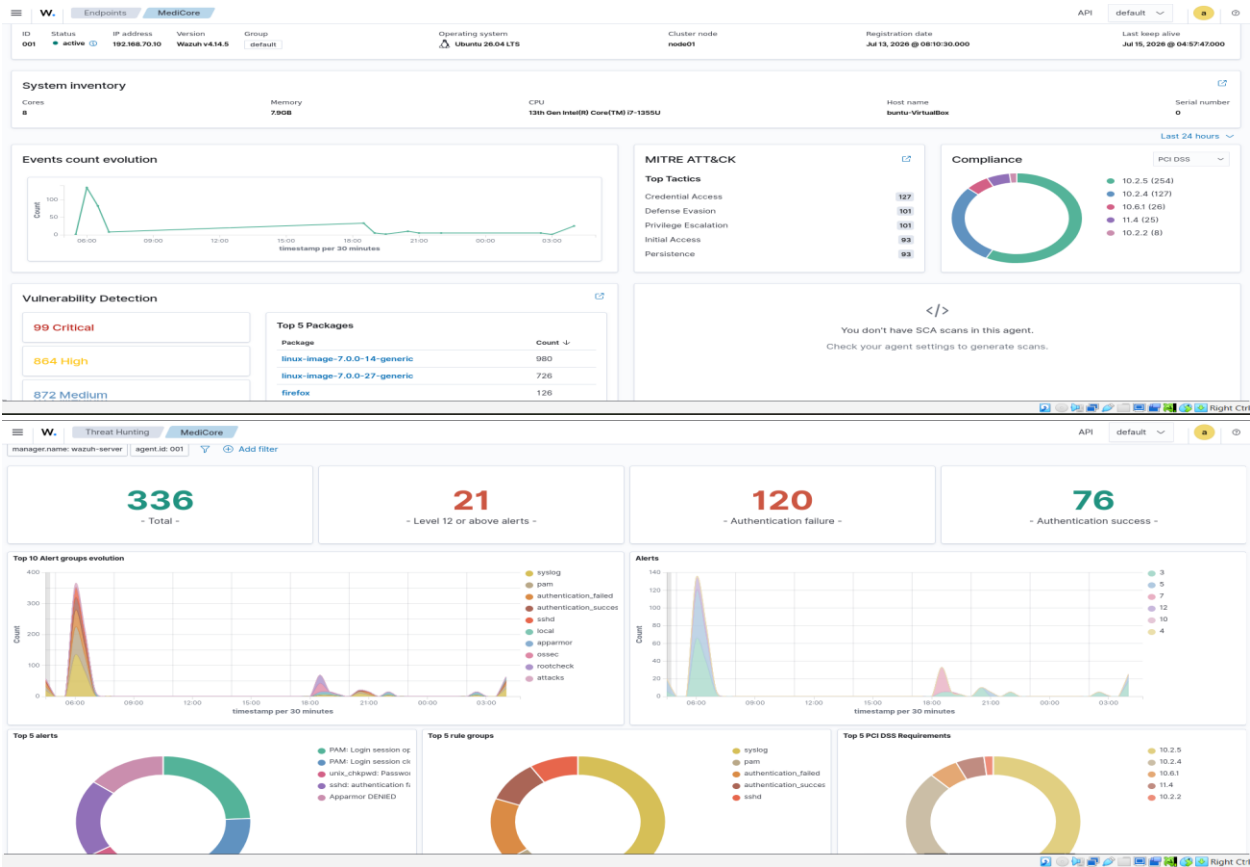
9.2.c Raw alert logs (timestamps, severity, rule ID)



Multiple authentication failures followed by a success. [View alerts of this Rule](#)

Information				
ID	Level	File	Path	
40112	12	0280-attack_rules.xml	ruleset/rules	
Groups				
syslog, attacks				
Details				
Timeframe	If_group	If_matched_group	Same_source_ip	
240	authentication_success	authentication_failures	true	
Compliance				
PCI DSS	GPG 13	GDPR	HIPAA	
10.2.4, 10.2.5, 11.4	7.1, 7.8	IV_35.7.d, IV_32.2	164.312.b	
NIST-800-53	TSC	MITRE Techniques	MITRE Tactics	
AU.14, AC.7, SI.4	CC6.1, CC6.8, CC7.2, CC7.3	Valid Accounts, Brute Force	Persistence,Privilege Escalation,Defense Evasion,Initial Access,Credential Access	
Related rules				
ID ↑	Description	Groups	Compliance	Level File
40101	System user successfully logged to the system.	invalid_login, syslog, attacks	PCI_DSS MITRE GPG13 HIPAA GDPR NIST_800_53 TSC	12 0280-attack_rules.xml
40102	Buffer overflow attack on rpc.statd	exploit_attempt, syslog, attacks	PCI_DSS GDPR NIST_800_53 TSC MITRE	14 0280-attack_rules.xml
		exploit_attempt		

9.3.c Correlation notes showing how alerts map to network evidence



9.4.c Any active response logs (block events, scripts run)

Firewall / Rules / LAN

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

Floating

WAN

LAN

DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/768 B	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	✗ 0/0 B	IPv4 *	LAN address	*	DMZ address	*	*	none			
☐	✓ 6/1.03 GiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	✓ 0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

↑ Add

↓ Add

Delete

Toggle

Copy

Save

Separator

Firewall / Rules / Edit

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LAN address

Source Address

/

Destination

Destination

☐ Invert match

DMZ subnets

Destination Address

/

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.



SNMP Daemon

Enable ☐ Enable the SNMP Daemon and its controls

SNMP Daemon Settings

Polling Port

Enter the port to accept polling events on (default 161).

System Location

System Contact

Read Community String

The community string is like a password, restricting access to querying SNMP to hosts knowing the community string. Use a strong value here to protect from unauthorized information disclosure.

SNMP Traps Enable

Enable ☐ Enable the SNMP Trap and its controls

SNMP Modules

SNMP modules

- ☒ MibII
- ☒ Netgraph
- ☒ PF
- ☒ Host Resources
- ☒ UCD
- ☒ Regex

Interface Binding

Internet Protocol

Bind Interfaces

- ☒ All
- ☐ WAN
- ☐ LAN
- ☐ DMZ

```

237
238 <ossec_config>
239 <active-response>
240   <disabled>no</disabled>
241   <command>firewall-drop</command>
242   <location>local</location>
243   <rules_group>5763</rules_group>
244   <timeout>0</timeout>
245 </active-response>
246 </ossec_config>
247

```

```

(kali@kali)-[~]
└─$ ssh admin@192.168.70.10
admin@192.168.70.10's password:
Permission denied, please try again.
admin@192.168.70.10's password:
Permission denied, please try again.
admin@192.168.70.10's password:
ssh_dispatch_run_fatal: Connection to 192.168.70.10 port 22: Connection timed out
Unable to fetch some archives, maybe run apt update or try with --fix-missing

(kali@kali)-[~]
└─$ ssh admin@192.168.70.10
admin@192.168.70.10's password:
Permission denied, please try again.
admin@192.168.70.10's password:
Permission denied, please try again.
admin@192.168.70.10's password:
admin@192.168.70.10: Permission denied (publickey,password).

```

10. KEY ANALYTIC QUESTIONS

1. ***Assets by Zone:***
 - a. *WAN:* The Internet.
 - b. *LAN:* Clinical and administrative workstations, SOC workstations, Wazuh Server, and medical devices (IoMT).
 - c. *DMZ:* Patient Portal and Telehealth Server.
2. ***Allowed Traffic:*** Essential traffic such as authorized EHR access for clinical staff and patient access to the telehealth portal via the DMZ.
3. ***Blocked Traffic:*** All direct traffic from the WAN to the LAN; unauthorized lateral traffic between the DMZ and internal clinical databases.
4. ***Identifying Suspicious Traffic:*** By monitoring unusual port scans, repeated authentication failures, and traffic spikes between internal zones that do not follow established baselines.
5. ***Business Risks of Non-Segmentation:*** Increased risk of lateral movement by attackers, potential HIPAA violations, data breaches of PHI, and disruption of life-saving clinical services.
6. ***Additional Controls:*** Implementation of Multi-Factor Authentication (MFA), regular vulnerability scanning, and automated "Active Response" in the SIEM to block malicious IPs instantly

❖ Final Thoughts....

THINKING SO MUCH ABOUT SOC ANALYSIS CLASS

Every alert has a story. Every log has a lesson.
In class, we don't just learn tools – we build a mindset.
We see the attack. We follow the evidence.
We protect the mission. 🛡️

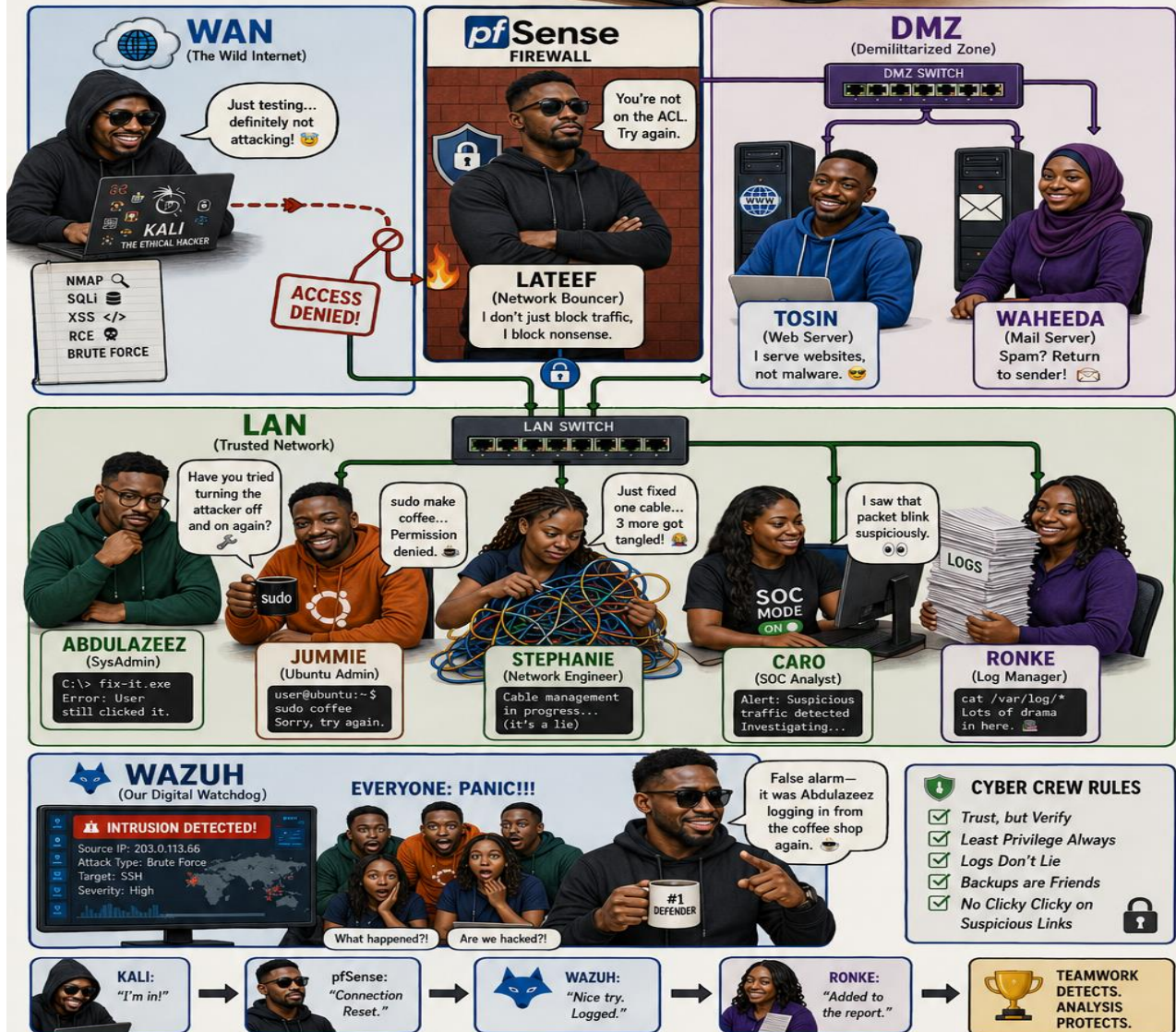
LEARNING IS BETTER WHEN WE BUILD TOGETHER!

So my amazing class team and I built a network topology inspired by real-life cybersecurity scenarios (with a little bit of humour 😊).

In cybersecurity, we don't guess.
We investigate.
We analyze.
We defend.

TODAY'S MISSION:

- ✅ Detect the bad guys
- ✅ Protect the network
- ✅ Analyze the evidence
- ✅ Respond like pros
- ✅ Have fun while doing it! 😊



Behind every alert is a story. Behind every analyst is a team.

Grateful for this class, this team, and the mindset we're building together. 💙 🧑‍💻 🛡️

#SOCAnalystInProgress #BlueTeam

#CyberCrew #Wazuh #pfSense #NetworkSecurity